Network Science

Lecture 03: Strong and Weak Ties

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1 April 2026

This lecture connects local friendship structure to global information flow.

Speaker notes

Pedagogical approach for this lecture:

- Use inquiry-first teaching throughout. Every module opens with a guiding question.
- This lecture builds on Lecture 02 (graph theory) and introduces the first sociological concepts.
- The key insight: local structure (triads, tie strength) has global consequences (information flow, bridging).
- Each module follows: Question → Tension → Interpretation → Stabilization.

Teaching goal for today:

- students should understand triadic closure and the clustering coefficient
- students should distinguish bridges from local bridges
- students should see why weak ties matter structurally
- students should be able to interpret empirical evidence from phone and social-media data

Strong and Weak Ties

This lecture connects local friendship structure to global information flow.

Focus

- triadic closure and strong triadic closure
- bridges and local bridges
- why weak ties matter for global connectivity
- empirical evidence from phone and social-media data

Module 1

Triadic Closure

> **Guiding Question:** Why do friends-of-friends so often become linked, and what does that imply for network structure?

Speaker notes

Hold this question open. Do not answer it immediately. Let students offer explanations — opportunity, trust, social pressure — before moving to the formal slide.

Module 1

Triadic Closure

> **Guiding Question:** Why do friends-of-friends so often become linked, and what does that imply for network structure?

$$\text{shared friend} \Rightarrow \text{increased probability of closure}$$

Triadic Closure

An **open triad** has two edges and one missing edge. **Triadic closure** is the process by which the missing edge appears over time.

> *Consider three nodes A, B, C where A–B and A–C exist but B–C does not. Triadic closure predicts B–C will form.*

Ask: before we explain why, what mechanisms could drive the missing edge to appear?

- Shared friends create environments to meet.
- Trust can be borrowed through a common contact.
- Social friction rewards closing the triangle to resolve inconsistency.

Speaker notes

The fragment reveals the three mechanisms after the question is posed in the slide body. This combines what were previously two separate slides (definition + "Why Does This Happen?") into one coherent question-then-answer slide.

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Ask: before we explain why, what mechanisms could drive the missing edge to appear?

- **Opportunity:** Shared friends create environments to meet.
- **Trust:** Trust can be borrowed through a common contact.
- **Incentive:** Social friction rewards closing the triangle to resolve inconsistency.

Strong Triadic Closure

> *If A has **strong** ties to both B and C, and B–C does not exist, then A violates the Strong Triadic Closure Property.*

Strong Triadic Closure Property: If a node maintains strong ties to two distinct neighbors, there must be at least a weak tie between those neighbors.

Speaker notes

Contrast the two cases: in the general case (open triad), any tie can close the triangle. In the strong case, the closing tie may be weak — but it must exist. The key distinction is that "strong" is a social label, not a graph-theoretic one.

Measuring Closure: Clustering Coefficient

What fraction of a node's potential triangles are actually closed?

Speaker notes

Math First, Then Meaning. Show the formula first, then ask students: What does $C_i = 1$ mean for a person's social circle? What does $C_i = 0$ mean? Could a person with $C_i = 0$ be a structural broker?

Measuring Closure: Clustering Coefficient

What fraction of a node's potential triangles are actually closed?

$$C_i = \frac{\text{number of edges among } i\text{'s neighbors}}{\binom{k_i}{2}}$$

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$$C_i = \frac{\text{number of edges among } i\text{'s neighbors}}{\binom{k_i}{2}}$$

where k_i is the degree of node i. Values range from 0 (no triangle) to 1 (complete neighborhood).

Quick Check 3.A: Clustering Coefficient

Node B has 4 neighbors: A, C, D, E. Among B's neighbors, exactly 2 edges exist: (A, C) and (C, D).

1. Compute C_B.
2. Is B's neighborhood highly clustered or sparse? What would that imply about B's social position?

Speaker notes

Give them 4 minutes — the second question is the important one. The numeric answer is $C_B = 2/6 = 1/3$. The interpretation is that B sits in a moderately sparse group, connecting nodes that don't all know each other. B could potentially act as a local broker between A (who knows C but not D/E) and D/E.

Quick Check 3.A: Solution

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2. Interpretation:

Quick Check 3.A: Solution

1. Compute C_B:

$$\binom{4}{2} = 6 \text{ possible edges}, \quad 2 \text{ actual edges} \implies C_B = \frac{2}{6} = \frac{1}{3}$$

2. Interpretation:

$C_B = 1/3$ indicates a sparse but not isolated neighborhood. B connects nodes that do not all know each other — a position with potential brokerage value, since B mediates between sub-clusters A, C and D.

Takeaway

> Triadic closure is a local mechanism, but repeated across a network it explains why social graphs become clustered — and why those clusters can be structurally separated by rare long-range ties.

Module 2

Bridges and Local Bridges

> **Guiding Question:** Which edges matter most for keeping information flowing across a network?

Speaker notes

Let the gap between "important" and "strongest" sit for a moment. Students likely equate strength with importance. The rest of this module builds the case that structural position matters more.

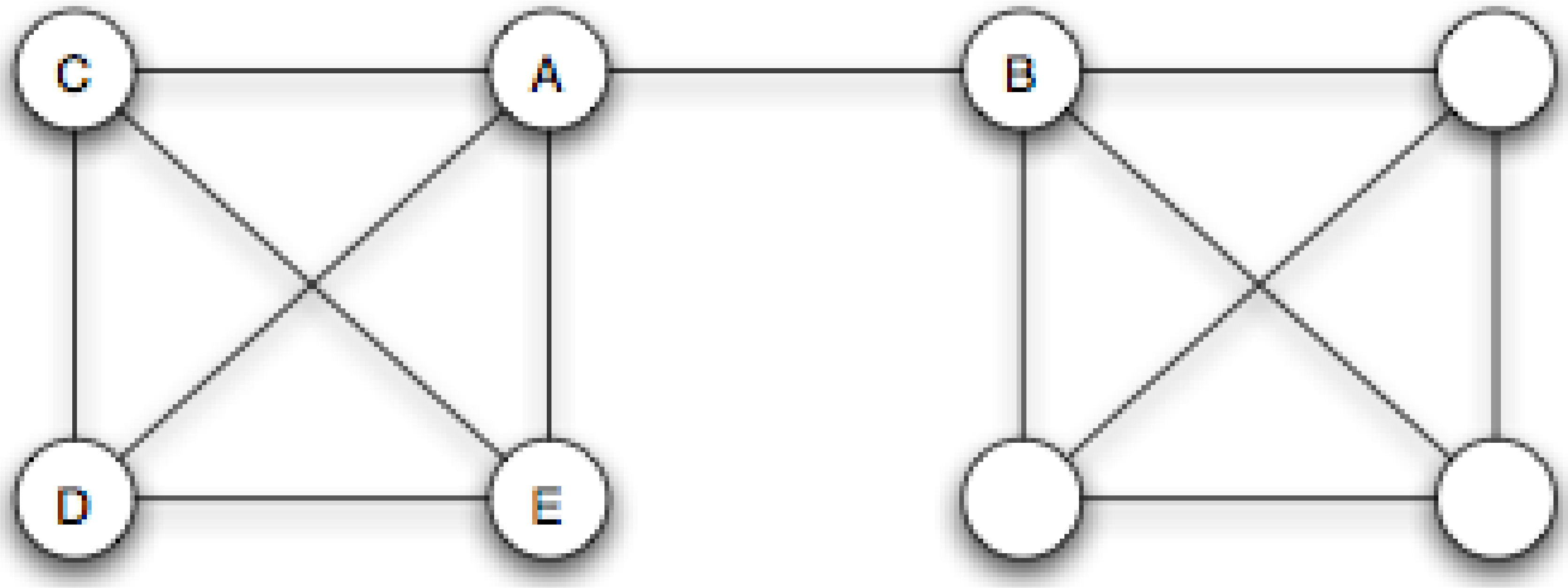
Module 2

Bridges and Local Bridges

> **Guiding Question:** Which edges matter most for keeping information flowing across a network?

$$\text{important edge} \neq \text{strongest edge}$$

Bridge



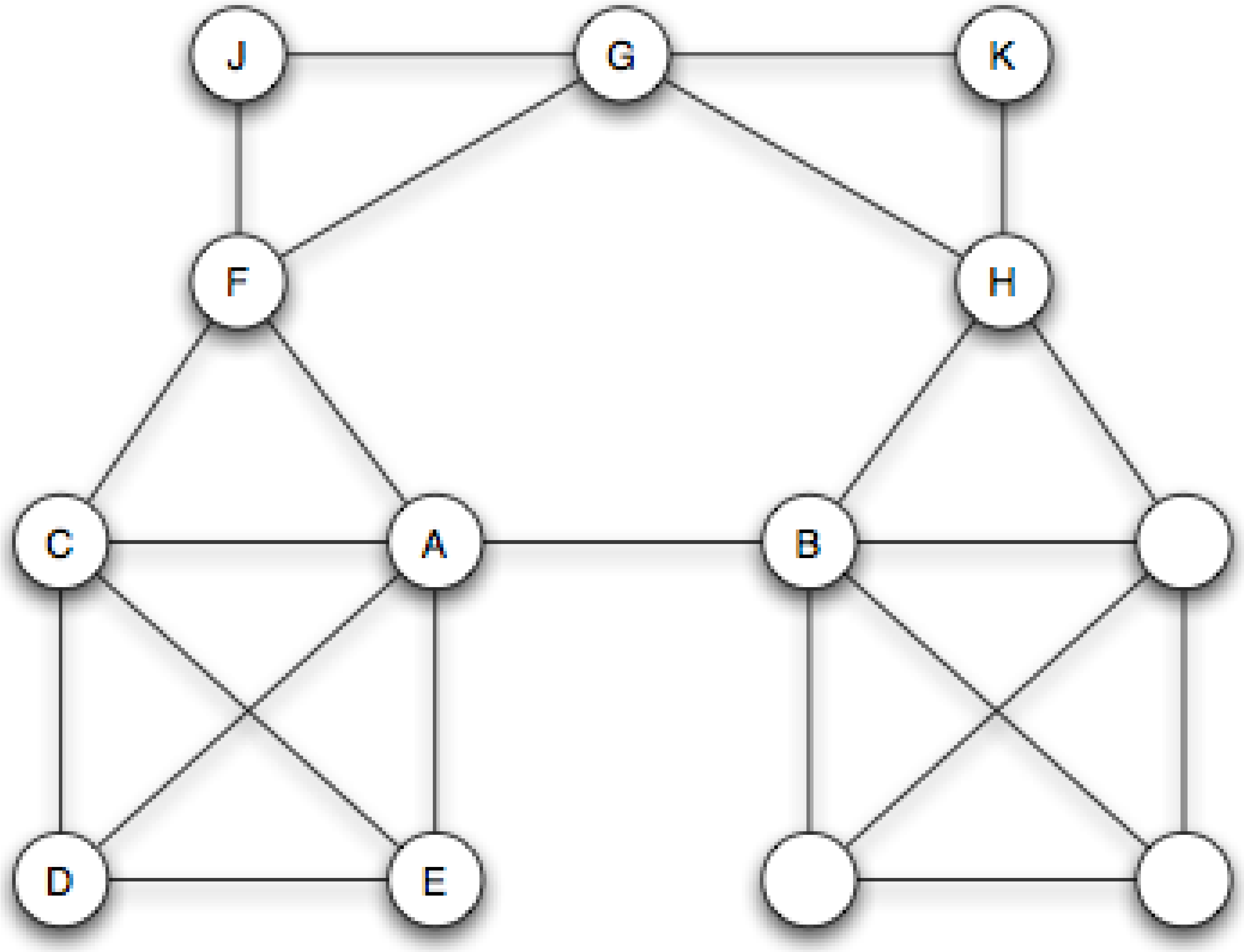
Source: Easley & Kleinberg 2010

A **bridge** is an edge whose removal increases the number of connected components. Equivalently: a bridge lies on no cycle.

Speaker notes

Ask students to verify the bridge condition: does the bridge edge lie on any cycle? No — removing it disconnects the graph. Ask: what happens to the distance between the endpoints after removal? It becomes infinite.

Local Bridge



Source: Easley & Kleinberg 2010

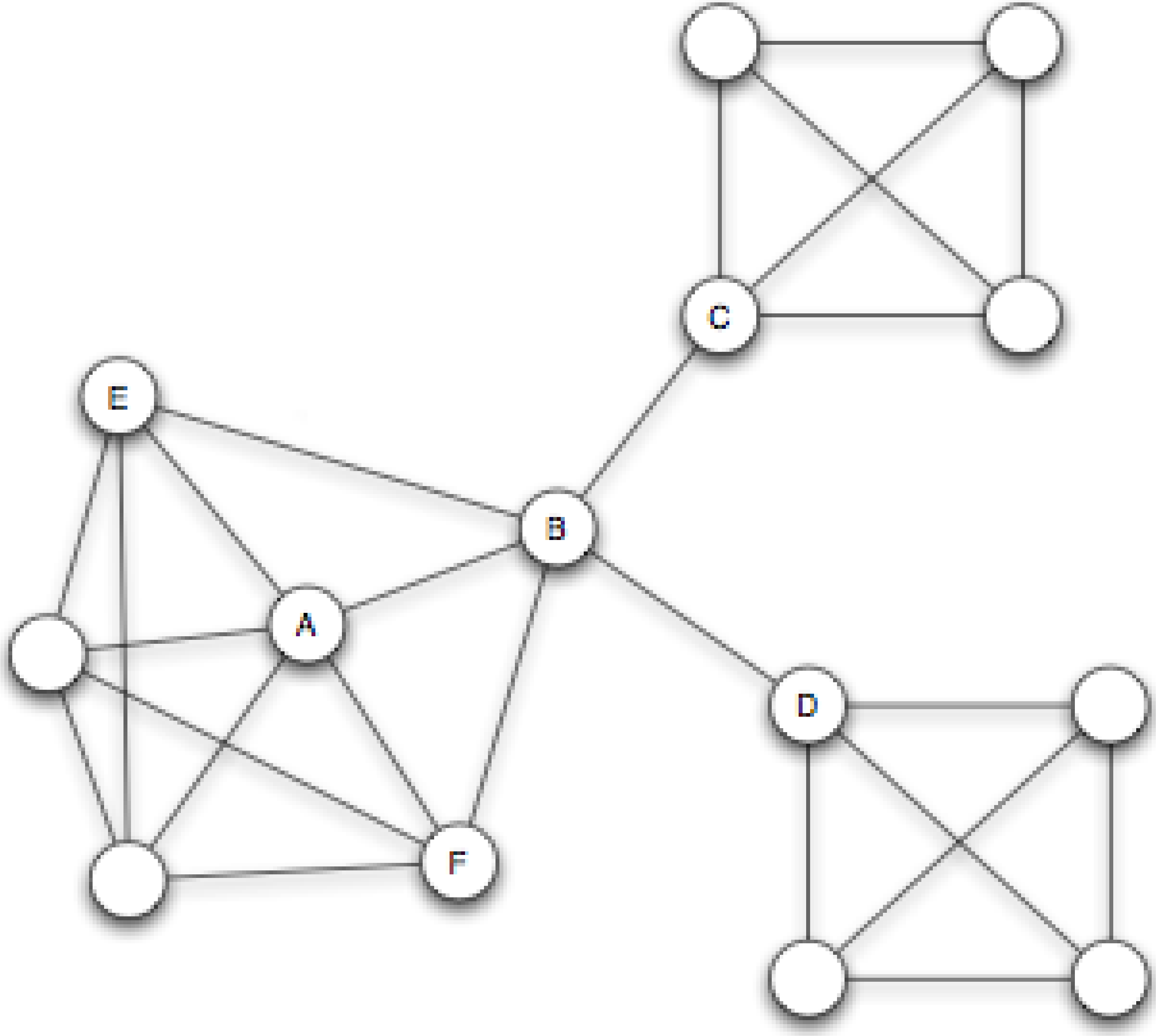
A **local bridge** links two nodes that share **no** neighbors. Removing it increases the distance between its endpoints to more than two.

Speaker notes

Contrast with a true bridge: a local bridge need not disconnect the graph. The endpoints could still reach each other via a long path — but that path goes through many hops.

Structural Holes and Social Capital

A node that acts as the sole link between otherwise disconnected groups spans a **structural hole** (Burt, 1992).



Source: Burt 1992

Why does this position matter?

- The broker receives early, non-redundant information from independent groups.
- The broker mediates any flow between groups that cannot communicate directly.

Speaker notes

Connect graph structure to social behavior. Ask: what happens to the Broker if triadic closure reconnects its contacts? The structural hole closes and the broker's monopoly evaporates. This is why individuals in brokering positions sometimes actively discourage their contacts from connecting directly.

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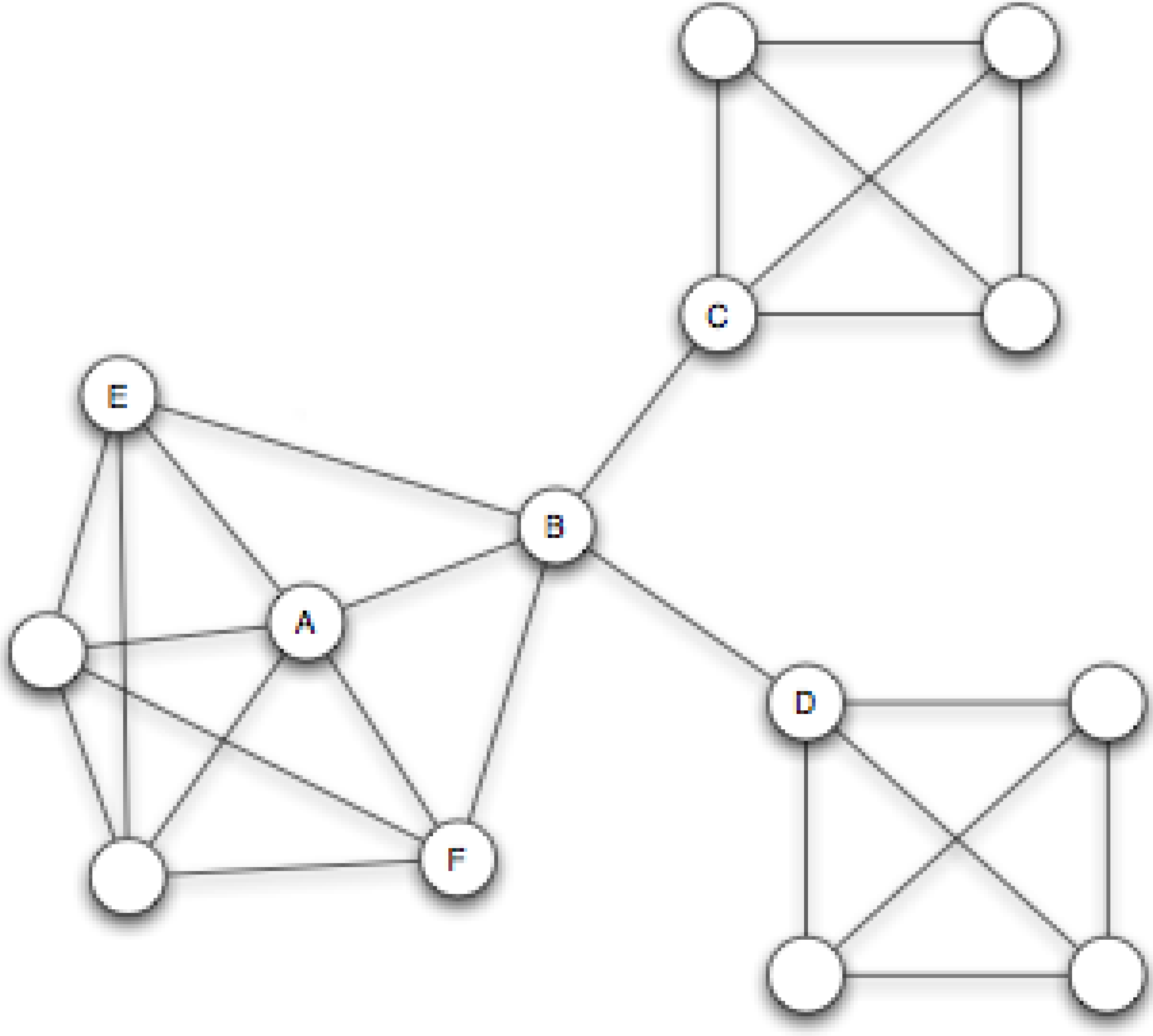

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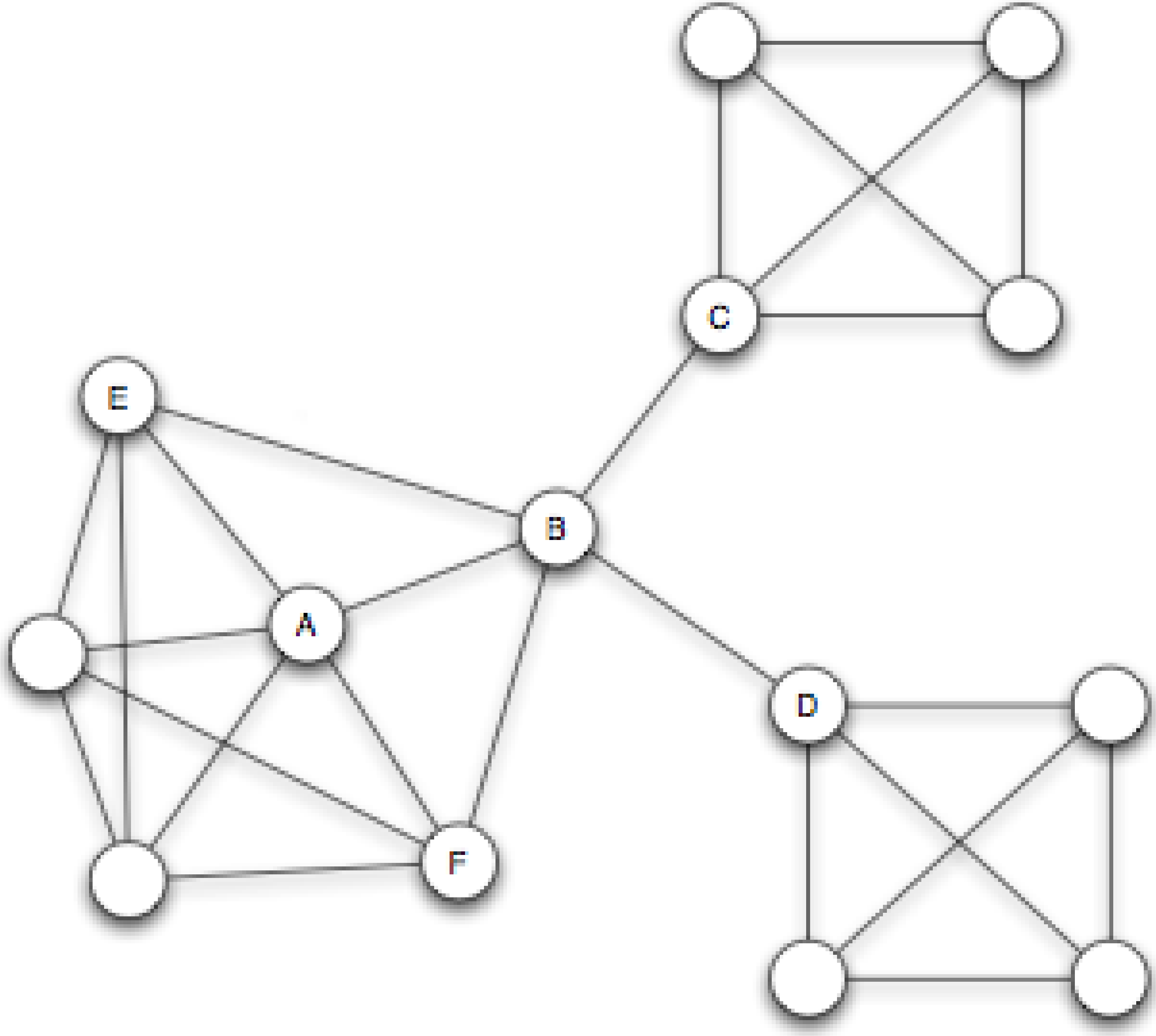

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Quick Check 3.B: Bridges and Brokers

The graph below represents a small research community. Each letter is a researcher.

> *A — B — C — D — E, plus edges B–C and C–D forming triangles.*

1. Which edges in this graph are **true bridges**?
2. Your advisor says "C is a broker". Do you agree? Justify using the definitions of local bridge and structural hole.

Speaker notes

Give them 4 minutes. The graph is: A–B, B–C, C–D, D–E, with additional edges B–C (already there) and C–D. So A–B is a bridge (removing it disconnects A). D–E is a bridge (removing it disconnects E). B–C and C–D are NOT bridges because they lie on cycles (triangle B–C–D). C is not a broker in the structural-hole sense — it sits inside a densely connected triangle, not between disjoint groups.

Quick Check 3.B: Solution

1. True bridges: Edges **A–B** and are bridges — removing either disconnects a node from the rest of the graph. Edges B–C and C–D lie on cycles and are not bridges.

No. C sits inside the triangle B–C–D, meaning B and D are already connected. C spans no structural hole: all of C's neighbors already know each other. A true broker would connect groups with no direct path between them.

Quick Check 3.B: Solution

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Quick Check 3.B: Solution

1. True bridges: Edges **A–B** and **D–E** are bridges — removing either disconnects a node from the rest of the graph. Edges B–C and C–D lie on cycles and are not bridges.

2. Is C a broker? No. C sits inside the triangle B–C–D, meaning B and D are already connected. C spans no structural hole: all of C's neighbors already know each other. A true broker would connect groups with no direct path between them.

Takeaway

> Structurally crucial edges are identified by position, not by weight. A node spanning structural holes gains information and control advantages — but loses them the moment triadic closure reconnects its contacts.

Module 3

Weak Ties as Structural Connectors

> **Guiding Question:** Why should local bridges tend to be weak ties rather than strong ties?

Speaker notes

Let students sit with this claim before moving on. It sounds counterintuitive — bridges feel important, so why would they be *weak*? The rest of this module builds the structural argument.

Module 3

Weak Ties as Structural Connectors

Guiding Question: Why should local bridges tend to be weak ties rather than strong ties?

$$\text{local bridge} + \text{strong triadic closure} \Rightarrow \text{weak tie}$$

Theorem Statement

Under strong triadic closure, every local bridge must be a weak tie.

Speaker notes

This is the compact formal form. Resist giving the proof immediately. Ask students: does this theorem say strong local bridges are impossible? (Answer: No — it says they cannot coexist with strong triadic closure. Real networks do violate STC, so the theorem is a conditional, not an absolute claim.)

Why Might This Be True?

Suppose A–D was a **strong** tie and also a local bridge. Then by strong triadic closure, A's other strong neighbors (B, C) must each have a tie to D. But then D shares neighbors with A — contradicting the local-bridge condition.

> *The assumption "A–D is strong AND a local bridge" forces a neighbor of D to exist among A's friends, destroying the local-bridge condition.*

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Speaker notes

Walk through this proof by contradiction carefully. The key move: the assumption forces a shared neighbor to exist, which destroys the local-bridge property. Ask students: what if only one of A's ties to B or C was strong? Does the contradiction still hold?

Granovetter: "Getting a Job" (1973)

Granovetter interviewed Boston-area residents who had recently found a new job through a social contact.

Is it the people we see most often who provide the most useful job leads?

Most job leads came from acquaintances seen rarely — not from close friends. The contacts providing novel opportunities were on the social periphery, not the social core.

Speaker notes

Ground the theorem in an empirical result students can relate to. Strong friends know the same people and the same opportunities you do. Weak acquaintances have access to entirely different social worlds — exactly as the theorem predicts for information flow across local bridges.

Granovetter: "Getting a Job" (1973)

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Is it the people we see most often who provide the most useful job leads?

Result: Most job leads came from acquaintances seen rarely — not from close friends. The contacts providing novel opportunities were on the social periphery, not the social core.

Interpretation

The theorem links a local social concept (tie strength) to a global structural one (bridging separate communities).

Weak ties are the primary channels through which non-redundant information — new jobs, new ideas, different perspectives — reaches individuals across community boundaries.

Quick Check 3.C: Strong Ties and Local Bridges

A network has three nodes: A, B, C. A–B is a **strong** tie. A–C is a **strong** tie. There is no edge between B and C.

1. Does this network satisfy the **Strong Triadic Closure Property**?
2. Is the edge A–B a local bridge? Can you tell from the information given?

Speaker notes

Give them 3 minutes. (1) No: A has two strong ties to B and C, but B–C does not exist, violating STC. (2) A–B is a local bridge only if B and A share no neighbors. Here B and C are not connected, so B and C are not shared neighbors of A and B. Whether A–B is a local bridge depends on whether there are other nodes — we cannot fully determine it from three nodes alone.

Quick Check 3.C: Solution

No. A has strong ties to B and C, but B and C are not connected — this violates STC.

2. Local bridge status of A–B: In this three-node network, A and B share no common neighbors (C connects to A but not to B). So . But the Weak Tie Theorem does not apply here — its precondition (STC holds) is violated. The theorem makes no claim about what happens when STC fails.

Quick Check 3.C: Solution

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Quick Check 3.C: Solution

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2. Local bridge status of A–B: In this three-node network, A and B share no common neighbors (C connects to A but not to B). So **A–B is a local bridge in this graph**. But the Weak Tie Theorem does not apply here — its precondition (STC holds) is violated. The theorem makes no claim about what happens when STC fails.

Takeaway

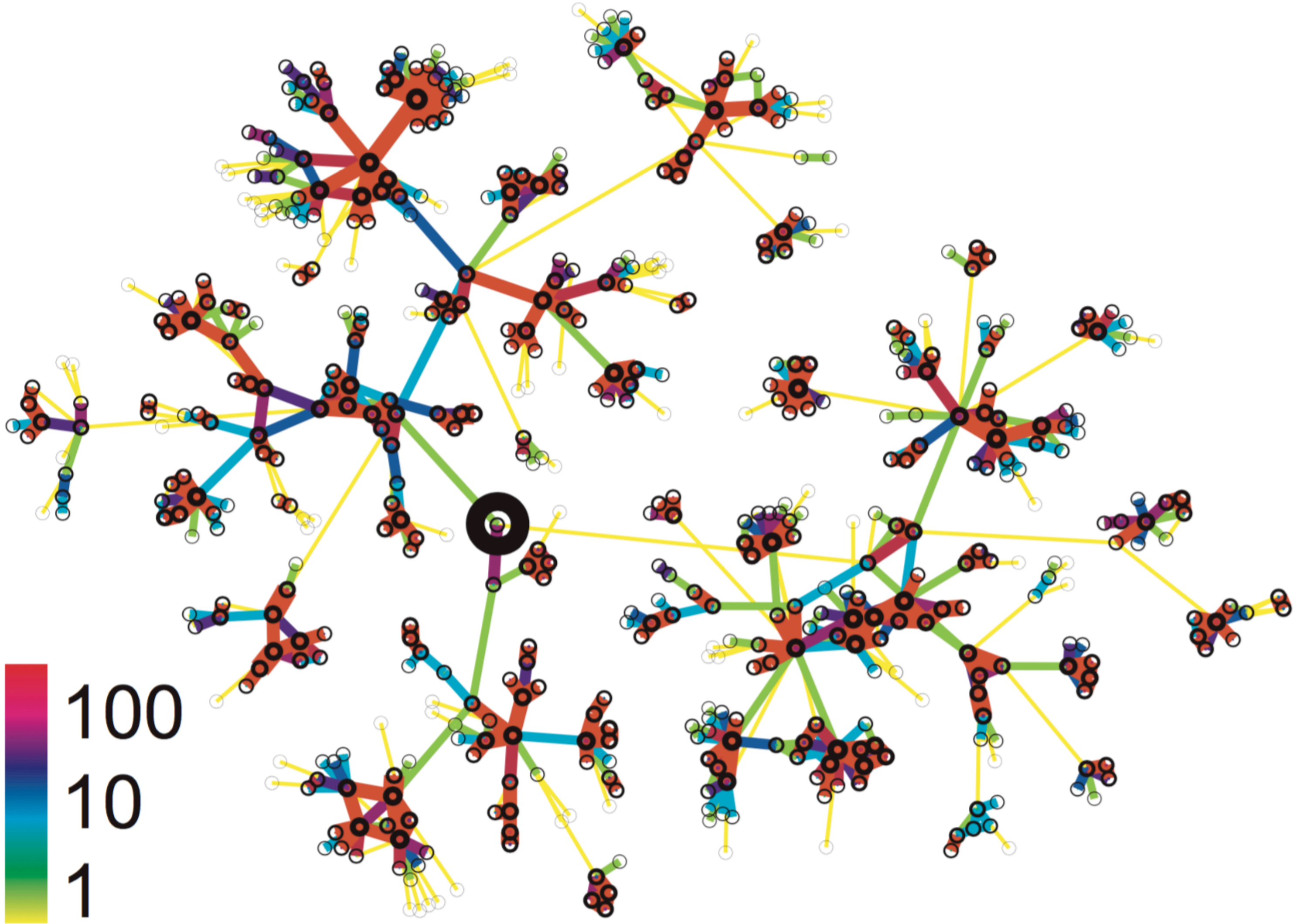
> Weak ties are structurally positioned as bridges between communities. The theorem explains *why* — not by sociology, but by graph structure: strong ties cluster locally, and local clustering destroys the bridge condition.

Module 4

Cell-Phone Evidence

> **Guiding Question:** Does the weak-ties theorem survive contact with real-world data at scale?

Setup: Modeling Communication as a Graph



Source: Onnela et al. 2007

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Onnela et al. (2007) modeled nationwide cell-phone call logs as an undirected weighted graph. Edge weight = total call duration between two numbers.

Speaker notes

Ask students: Is call duration a good proxy for tie strength? What gets lost? (Loss: directionality, content, context, relationship history before phones.) This is the modeling choice made visible.

Formalizing Almost-Local Bridges

In a real weighted graph, true local bridges are rare. Instead, we measure **Neighborhood Overlap**:

$$O(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}$$

where $N(u)$ is the set of neighbors of u, excluding v (and vice versa).

What does $O(u, v) = 0$ mean?

The two endpoints share common neighbors — the edge is a perfect local bridge.

Speaker notes

Math First, Then Meaning. Show the formula, ask for its value at the extremes. Note: this is the Jaccard similarity coefficient applied to neighbor sets. $O = 1$ means all neighbors are shared (deep within a clique). $O = 0$ means the edge is a local bridge.

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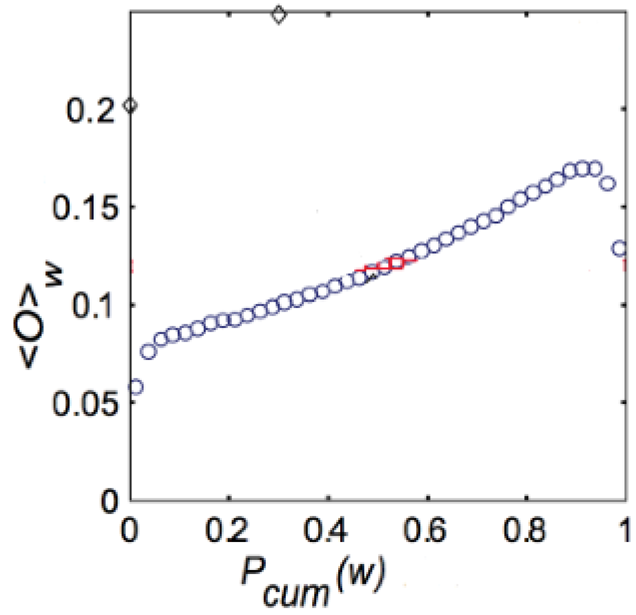
Evidence: Tie Strength vs. Overlap



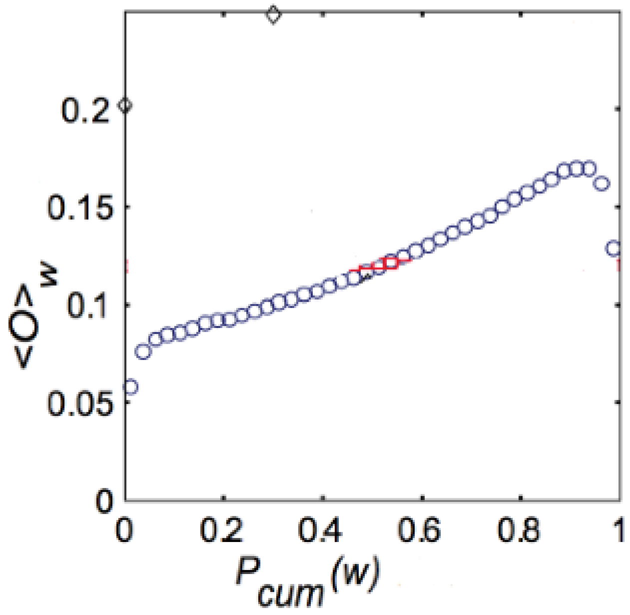
Source: Onnela et al. 2007

Does tie strength predict neighborhood overlap?

Speaker notes

Ask students to read the axes before revealing the fragment: the X-axis is call duration rank (percentile), the Y-axis is neighborhood overlap. What pattern do you expect? Then reveal.

Evidence: Tie Strength vs. Overlap



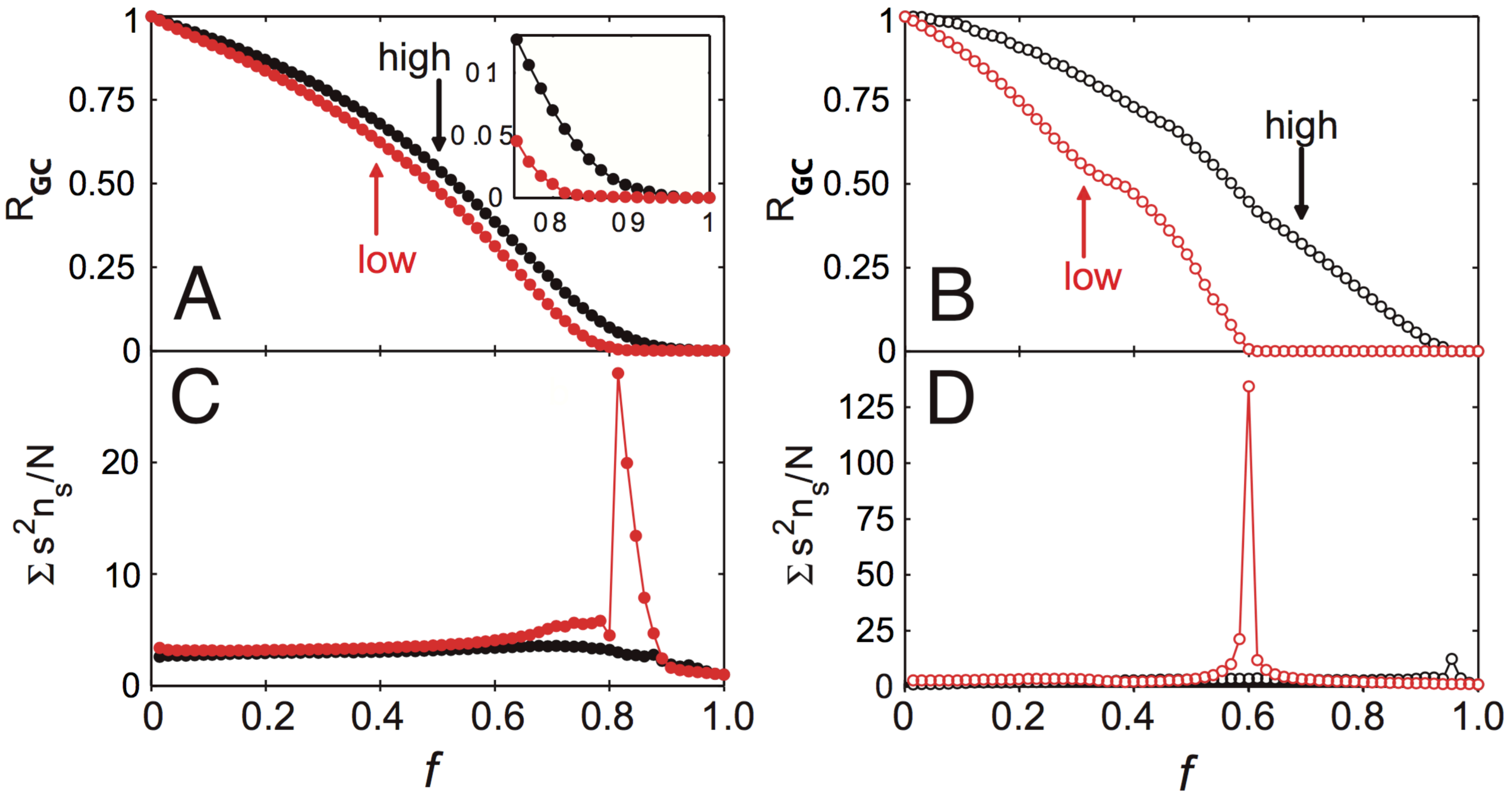
Source: Onnela et al. 2007

Does tie strength predict neighborhood overlap? Yes: weak ties (low strength percentile) cluster near

$$O \approx 0$$

. Strong ties accumulate at higher overlap values — consistent with Granovetter's prediction.

Evidence: Network Robustness



Source: Onnela et al. 2007

Which edge removal strategy breaks the network fastest?

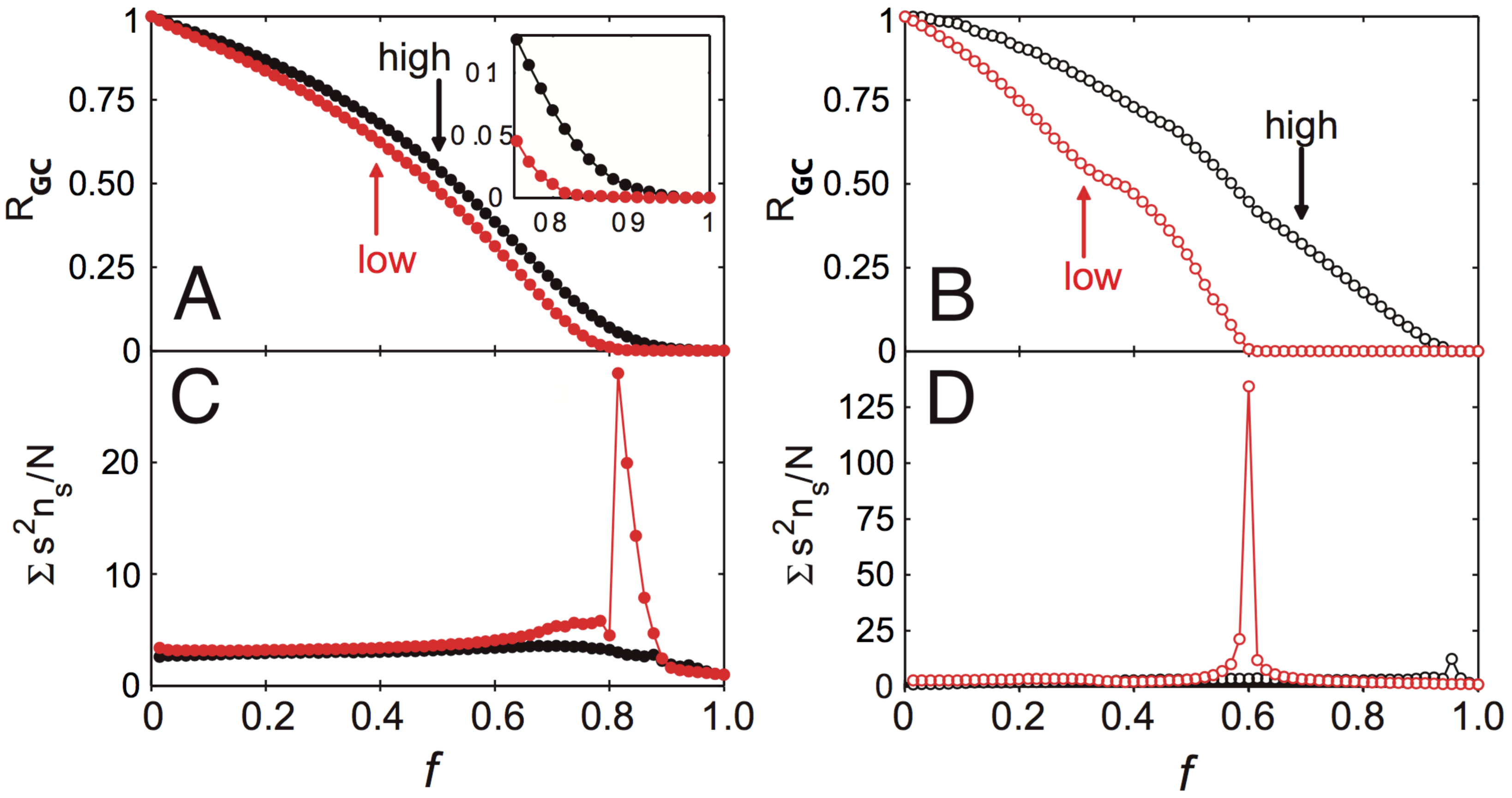
Removing edges in

tie-strength order (weakest first) fragments the giant component far faster than removing strongest ties first.

Speaker notes

Ask: Why would removing ties that carry very few call-minutes break the network so dramatically? Answer: because those weak ties are the local bridges between communities. Remove the bridges, separate the components.

Evidence: Network Robustness



Source: Onnela et al. 2007

Which edge removal strategy breaks the network fastest?

Removing edges in **increasing** tie-strength order (weakest first) fragments the giant component far faster than removing strongest ties first.

Quick Check 3.D: Neighborhood Overlap

Nodes A and B are connected. Their neighbor sets (excluding each other) are:

- $N(A) = C, D, E$
- $N(B) = D, E, F$

1. Compute $O(A, B)$.
2. Is A–B closer to a local bridge or a clique edge? What does this imply about the information A would receive through B?

Speaker notes

Give them 4 minutes. $|N(A) \cap N(B)| = |D, E| = 2$. $|N(A) \cup N(B)| = |C, D, E, F| = 4$. $O = 2/4 = 0.5$. Interpretation: moderate overlap — neither a pure bridge nor deep inside a clique. A receives some non-redundant information from B (via F) but also redundant information (D, E already known to A).

Quick Check 3.D: Solution

A–B has moderate overlap. It is not a local bridge ($O > 0$), but nor is it embedded in a clique ($O < 1$). A receives partly redundant information through B. If O were near 0, B would be A's primary source of information from a distant community.

Quick Check 3.D: Solution

$$O(A, B) = \frac{|D, E|}{|C, D, E, F|} = \frac{2}{4} = 0.5$$

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Interpretation: A–B has moderate overlap. It is not a local bridge ($O > 0$), but nor is it embedded in a clique ($O < 1$). A receives partly redundant information through B. If O were near 0, B would be A's primary source of information from a distant community.

Takeaway

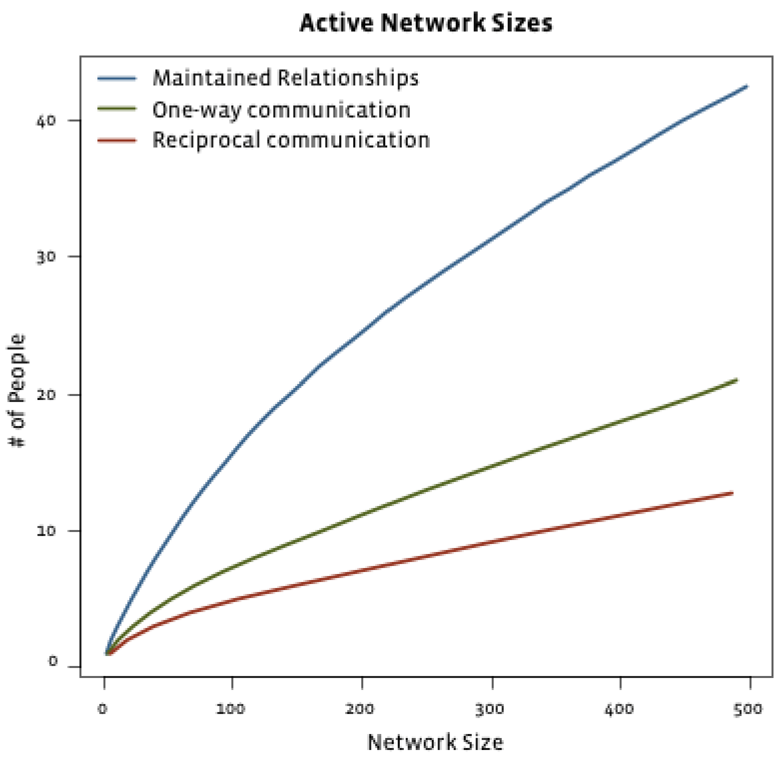
The weak-ties argument is not sociological speculation — it has measurable structural signatures at scale. Neighborhood overlap operationalizes the bridge concept, and empirical data confirm that weak ties carry it.

Module 5

Social Media

Guiding Question: How do strong and weak ties change when the network moves to online platform

Evidence: Facebook (2009)



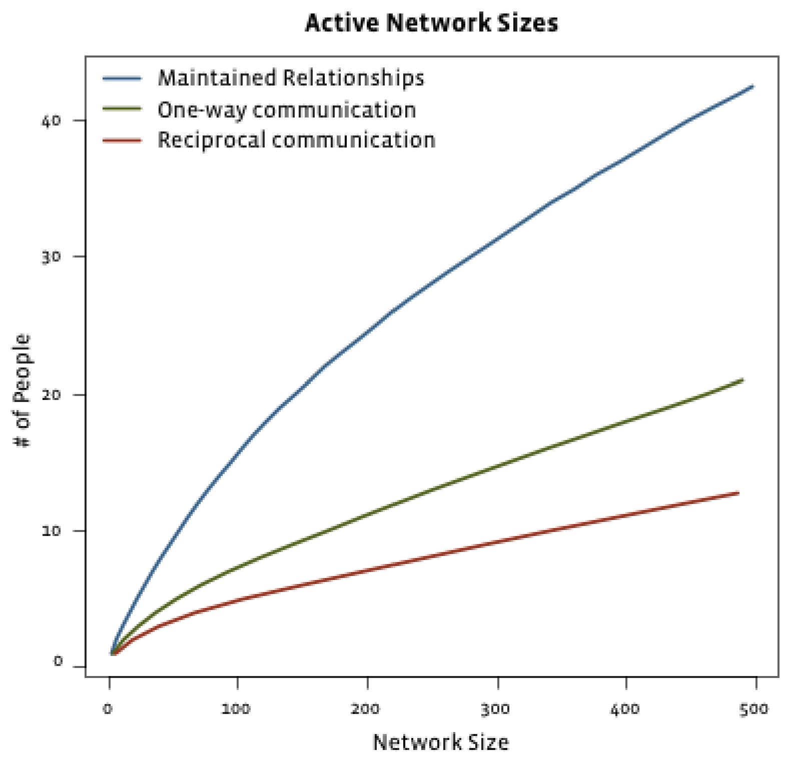
Source: Easley & Kleinberg 2010

As total friend count grows, what happens to actively maintained relationships?

Speaker notes

The figure shows that total friends grow rapidly with network size, while maintained and directed one-way contacts plateau. Cognitive limits still bind strong tie capacity — Dunbar's number (~150) applies even online. Point this out before revealing the fragment.

Evidence: Facebook (2009)

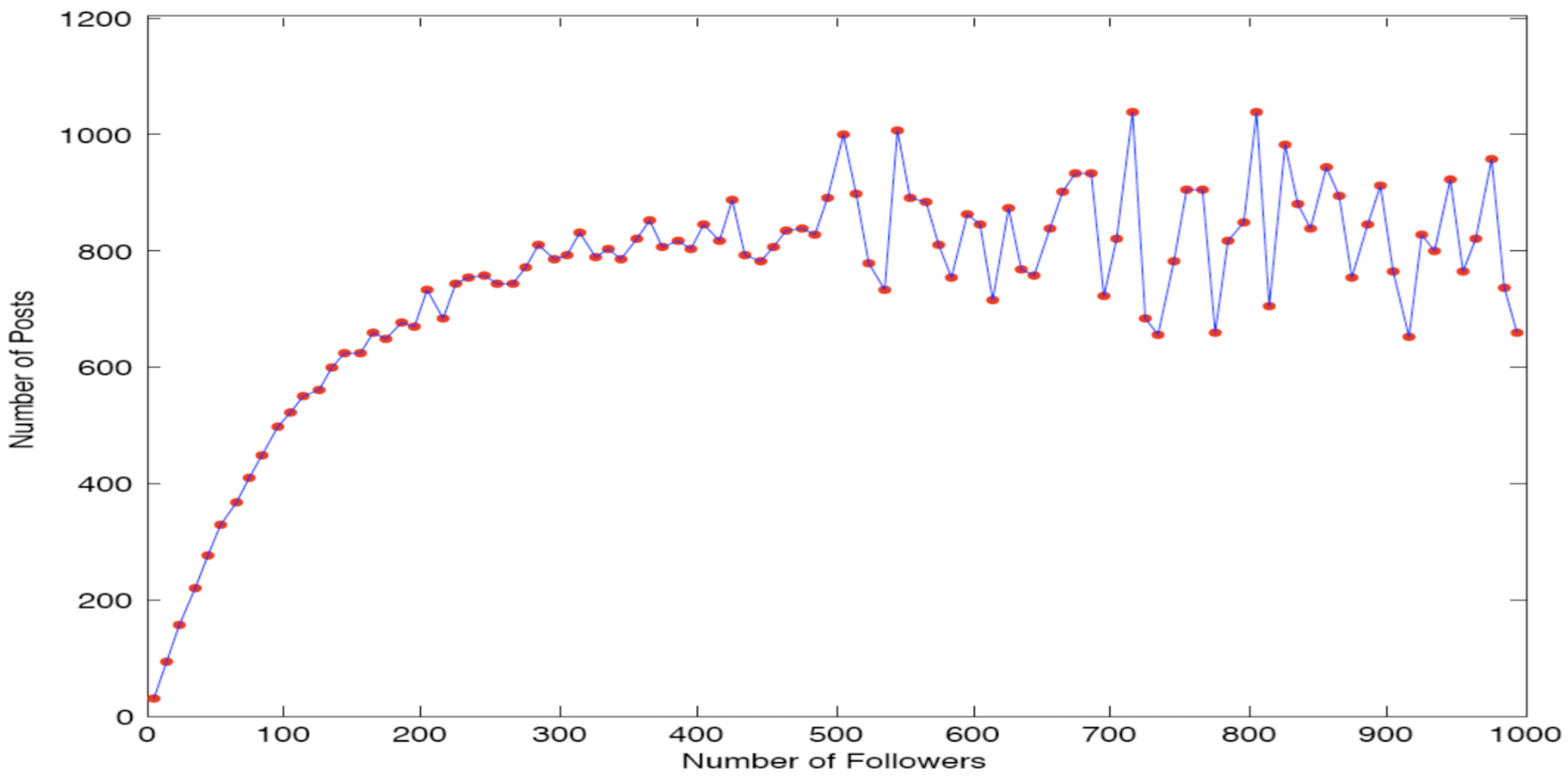


Source: Easley & Kleinberg 2010

As total friend count grows, what happens to actively maintained relationships?

Only a small, roughly constant number are "maintained" (reciprocal communication). The rest are passive weak ties.

Evidence: Twitter (Huberman et al.)



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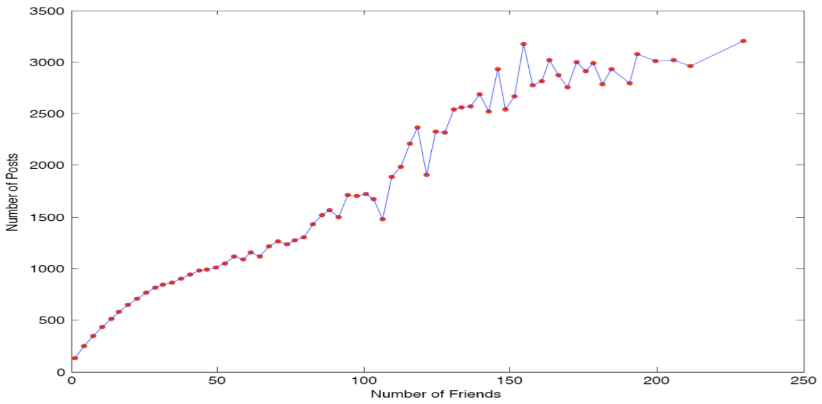

Source: Huberman et al. 2009

On asymmetric platforms, the *follower* count can reach millions (zero-cost weak ties), while *friends* — mutual contacts with whom one actually interacts — remain a small subset.

Speaker notes

Ask: why does the follower graph (left) scatter so much more than the friends graph (right)? The relationship between posting activity and follower count is much noisier than the one with actual friends. "Friends" here is the Twitter-era term for mutual follows with actual interaction.

Interpretation

Does social media collapse the distinction between weak and strong ties?

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Does social media collapse the distinction between weak and strong ties?

No — it amplifies it. Platforms remove the cost of maintaining weak ties (passive contact, one-click follow), but strong ties still require active, reciprocal effort. The result is a much wider weak-tie periphery surrounding an unchanged strong-tie core.

Quick Check 3.E: Dense vs. Sparse Follower Networks

Two students, X and Y, both use the same online platform. X follows 2000 accounts; Y follows 80 accounts but everyone follows everyone else.

1. Who is likely to have a higher average neighborhood overlap across their edges?
2. What does this imply about the kind of information X and Y each receive?

Speaker notes

Give them 4 minutes. For X: following 2000 accounts, many via recommendation algorithm, means few shared neighbors per edge — low average overlap, many weak-tie-like connections, access to diverse non-redundant information. For Y: 80 accounts in a dense mutual-follow cluster means high overlap — Y is embedded deeply in a clique, receiving mostly redundant information.

Quick Check 3.E: Solution

Y likely has average overlap. Y's 80 contacts form a dense mutual cluster — many shared neighbors per edge. X's 2000 contacts include many algorithm-driven follows with near-zero shared neighborhood.

- X receives information — the structural signature of many weak ties and local bridges.
- Y receives information — the structural signature of a dense clique with few connections outside.

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Quick Check 3.E: Solution

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Y likely has **higher** average overlap. Y's 80 contacts form a dense mutual cluster — many shared neighbors per edge. X's 2000 contacts include many algorithm-driven follows with near-zero shared neighborhood.

2. Information implications:

- X receives **diverse, non-redundant** information — the structural signature of many weak ties and local bridges.
- Y receives information — the structural signature of a dense clique with few connections outside.

Quick Check 3.E: Solution

1. Neighborhood overlap:

Y likely has **higher** average overlap. Y's 80 contacts form a dense mutual cluster — many shared neighbors per edge. X's 2000 contacts include many algorithm-driven follows with near-zero shared neighborhood.

2. Information implications:

- X receives **diverse, non-redundant** information — the structural signature of many weak ties and local bridges.
- Y receives **redundant, reinforced** information — the structural signature of a dense clique with few connections outside.

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Takeaway

> Social media does not remove the weak/strong-tie distinction — it stretches the weak-tie periphery enormously while leaving the strong-tie core bounded by human cognitive limits.

Wrap-Up

- local friendship structure creates predictable global effects
- weak ties are often not socially strongest, but structurally most valuable
- empirical network data broadly supports the theoretical pattern

Looking Ahead: Lecture 04

How do we identify groups and roles within a network? Which nodes are important — and by what definition?

Speaker notes

Plant the question for next time: importance is not one concept. Degree, closeness, betweenness, and eigenvector centrality each formalize a different intuition. Community detection asks a complementary question: where are the dense groups?

Image Credits & References

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